

Lab 2 — Applying AI in Healthcare Contexts

Course: ITAI 4376 — AI Applications

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Institution: Houston City College | Spring 2026

Overview

This lab explored how artificial intelligence is being applied in healthcare settings, with a focus on clinical decision support, predictive analytics, and the use of electronic health record data. The lab examined real-world use cases where machine learning models assist clinicians in making more informed decisions.

What I Learned

I gained an understanding of how healthcare data differs from general-purpose datasets — particularly in terms of complexity, privacy constraints, and the consequences of model errors in clinical contexts. I learned how AI tools are being used to predict patient outcomes, flag at-risk individuals, and assist with diagnostic workflows. The lab also reinforced why domain knowledge is essential when applying AI in high-stakes fields like medicine.

Challenges Faced

One of the more significant challenges was understanding the nature of healthcare data, which is often incomplete, inconsistently formatted, and subject to strict regulatory requirements such as HIPAA. Translating raw health data into a usable format for machine learning while preserving privacy and accuracy required careful consideration at every step.

Tools and Technologies Used

- Python 3
- Jupyter Notebook

- pandas
- scikit-learn
- Synthea (synthetic EHR data)

Key Takeaway

Applying AI in healthcare requires not only technical skill but also a deep respect for patient privacy, clinical accuracy, and the ethical responsibilities that come with building systems that influence medical decisions.